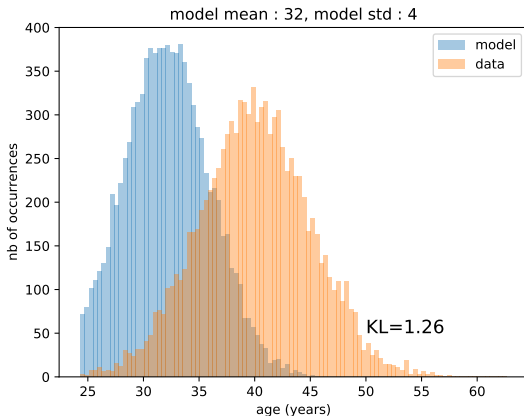


Machine learning Tek 5: density estimation



Maximum likelihood

KL divergence

Kernel density estimation

Density estimation

Objective : compute a probability distribution that represents the data well.

Maximum Likelihood

The **Maximum Likelihood** method is a famous method.

We observe a dataset $D_n = (x_1, \dots, x_n)$.

We first need to choose a **model** (which is the distribution) of the dataset, p .

Then, we must optimize the **parameters of this model**, noted θ .

Maximum Likelihood

The **likelihood** (vraisemblance) of the model is

$$L(\theta) = p(x_1, \dots, x_n | \theta) \quad (1)$$

Here, p denotes the probability of observing the sample $(x_1, \dots, x_n)$, when the model had the parameter θ , (or the probability density, in the corresponding context).

Maximum Likelihood

The **likelihood** (vraisemblance) of the model is

$$L(\theta) = p(x_1, \dots, x_n | \theta) \quad (2)$$

This is the function that we want to **maximize**, by choosing the best possible θ (optimization problem).

If $(x_1, \dots, x_n)$ are conditionally independant, then it writes :

$$L(\theta) = \prod_{i=1}^n p(x_i | \theta) \quad (3)$$

Remark on max-likelihood

Most of the time it's written this way : "minimise $-\log L(\theta)$ "

Because the log **transforms the product into a sum**, which is easier to **differentiate**.

$$-\log L(\theta) = -\sum_{i=1}^n \log(p(x_i|\theta)) \quad (4)$$

Example 1

Exercice 1 : We observe the data $(1, 0)$. We assume that these data come from a random variable that follows a Bernoulli distribution of parameter p . What is the likelihood of these observations as a function of p ?

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$$L = P(1|p)P(0|p) \quad (5)$$

For which value of p is this likelihood **maximum**?

Example 2

Exercise 2 : Same question when the data $(1, 0, 1)$ is observed.

Example 3

Exercise 3 : We observe the data $(2.5, 3.5)$. We assume that these data come from a normal law of parameters μ and σ .
What is the likelihood of (μ, σ) ?

Example 3

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$$\begin{aligned} L &= p(2.5|\mu, \sigma)p(3.5|\mu, \sigma) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{2.5-\mu}{\sigma}\right)^2} \times \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{3.5-\mu}{\sigma}\right)^2} \end{aligned} \quad (6)$$

Example 3

Exercice 3 : We observe the data (2.5, 3.5). We assume that these data come from a normal law of parameters μ and σ .

$$\begin{aligned} L &= p(2.5|\mu, \sigma)p(3.5|\mu, \sigma) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{2.5-\mu}{\sigma}\right)^2} \times \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{3.5-\mu}{\sigma}\right)^2} \end{aligned} \quad (7)$$

We want to show that the likelihood is maximum for :

- ▶ $\hat{\mu} = \frac{2.5+3.5}{2}$
- ▶ $\hat{\sigma}^2 = \frac{(2.5-\hat{\mu})^2 + (3.5-\hat{\mu})^2}{2}$

Kullback-Leibler Divergence

The KL divergence is a measure of the discrepancy between two **distributions**.

Expected value (espérance)

- For a discrete random variable X that takes the values x_i with probability p_i :

$$E(X) = \sum_{i=1}^n p_i x_i \quad (8)$$

- For a continuous random variable X with density $p(x)$:

$$E(X) = \int x p(x) dx \quad (9)$$

Kullbach-Leibler Divergence

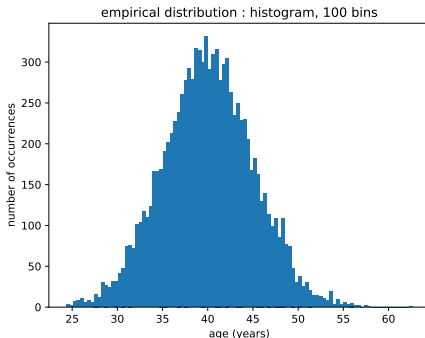
- ▶ The samples $(x_1, \dots, x_n)$ are described by an empirical distribution.
- ▶ The **Kullbach-Leibler divergence** is a tool to compare distributions.
- ▶ It is not a distance : it is not symmetric, no triangular inequality.

Exercise 3 : Fitting a distribution

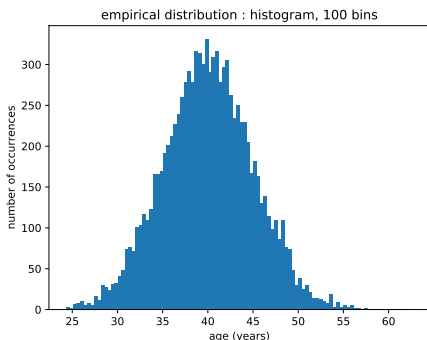
- ▶ `cd day_4/density_estimation/kl_divergence`
- ▶ A two dimensional dataset is contained in `empirical_distribution.csv`. It represents the **age distribution** of some groupe of people. We want to study this age distribution.
- ▶ Using `compute_kl.py`, we will find a model M such that $KL(M||\tilde{P})$ is small, with $\tilde{P}$ the empirical distribution of the data.

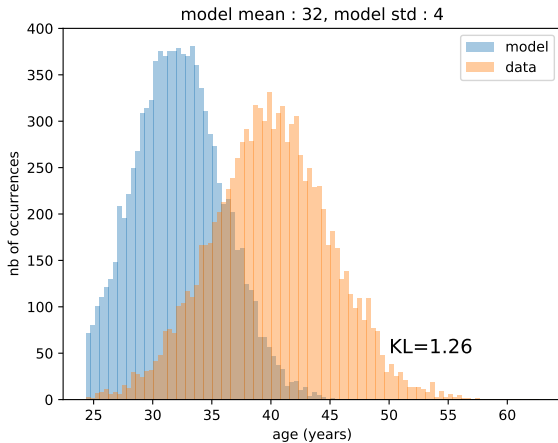
Exercise 3 : Fitting a distribution

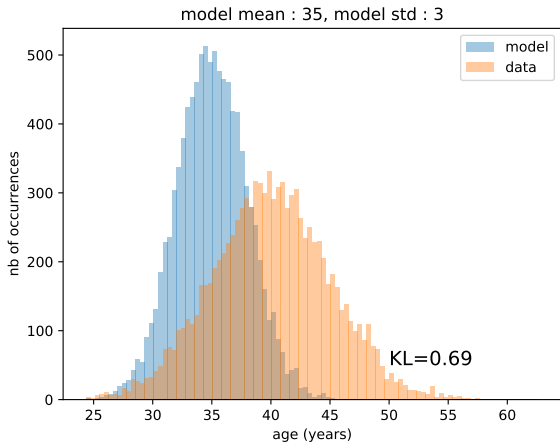
- **First step** : choice of the model
- Plot the histogram of the data : what model seems to be relevant ?

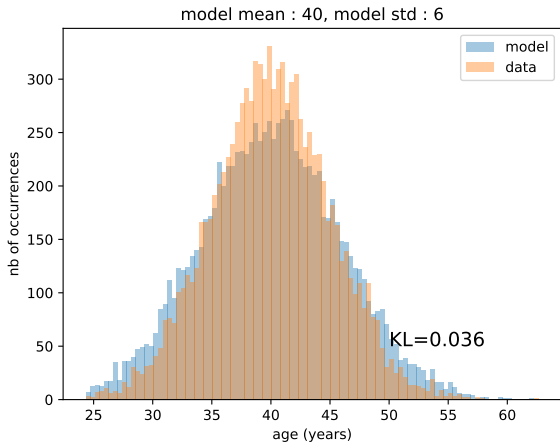


- ▶ We will use **normal laws**. We want to fit the normal law that is **the closest to the empirical data**
- ▶ We measure the proximity between the model and the empirical data with the KL divergence.









Kernel density estimation (non-parametric model)

https:

`//seaborn.pydata.org/generated/seaborn.kdeplot.html`

Example in **kde/**

